Exp2: Create a Docker image from a Docker file to run a simple web application.

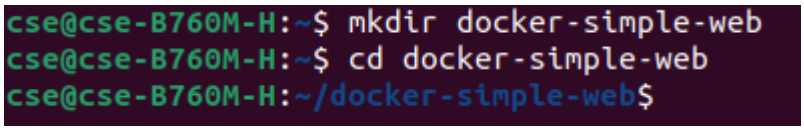
Software Requirements

- Ubuntu 22.04 LTS
- Docker Engine
- Python (inside container only)
- Web browser

Step 1: Create Project Directory

```
mkdir docker-simple-web

cd docker-simple-web
```



Step 2: Create a Simple Web Application

Create a file named **app.py**:

```
nano app.py
```

Type the following:

```
from flask import Flask

app = Flask(__name__)


@app.route("/")

def hello():

    return "Hello from Docker Web App!"


if __name__ == "__main__":

    app.run(host="0.0.0.0", port=5000)
```

```
  GNU nano 6.2
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello from Docker Web App!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Save and exit.

Step 3: Create requirements file

```
nano requirements.txt
```

Add:

```
flask
```

Save and exit.

Step 4: Create Dockerfile

```
nano Dockerfile
```

Type the following:

```
FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]
```

```
  GNU nano 6.2
FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]
```

Save and exit.

Step 5: Build Docker Image

```
docker build -t simple-web-app .
```

```
cse@cse-B760M-H:~/docker-simple-web$ docker build -t simple-web-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/7 : FROM python:3.10-slim
3.10-slim: Pulling from library/python
0c8d55a45c0d: Pulling fs layer
690eaffcf0e9: Pulling fs layer
d0d1977db485: Pulling fs layer
63ebb7511fc1: Pulling fs layer
63ebb7511fc1: Waiting
690eaffcf0e9: Download complete
d0d1977db485: Verifying Checksum
d0d1977db485: Download complete
0c8d55a45c0d: Verifying Checksum
0c8d55a45c0d: Download complete
0c8d55a45c0d: Pull complete
690eaffcf0e9: Pull complete
d0d1977db485: Pull complete
63ebb7511fc1: Verifying Checksum
63ebb7511fc1: Download complete
63ebb7511fc1: Pull complete
Digest: sha256:e508a34e5491225a76fbb9e0f43ebde1f691c6a689d096d7510cf7fb17d4ba6f
Status: Downloaded newer image for python:3.10-slim
 ---> 988b1bc80158
Step 2/7 : WORKDIR /app
 ---> Running in 785ff8ef48fb
 ---> Removed intermediate container 785ff8ef48fb
 ---> 7577978671de
Step 3/7 : COPY requirements.txt .
 ---> 69a3abbd6e58
Step 4/7 : RUN pip install -r requirements.txt
 ---> Running in 276824450c3a
Collecting flask
  Downloading flask-3.1.2-py3-none-any.whl (103 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.3/103.3 kB 5.1 MB/s eta 0:00:00
Collecting jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 kB 9.5 MB/s eta 0:00:00
Collecting blinker>=1.9.0
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting werkzeug>=3.1.0
  Downloading werkzeug-3.1.5-py3-none-any.whl (225 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 225.0/225.0 kB 24.8 MB/s eta 0:00:00
Collecting markupsafe>=2.1.1
  Downloading markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting click>=8.1.3
  Downloading click-8.3.1-py3-none-any.whl (108 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 108.3/108.3 kB 45.2 MB/s eta 0:00:00
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, flask
Successfully installed blinker-1.9.0 click-8.3.1 flask-3.1.2 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.5
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
 ---> Removed intermediate container 276824450c3a
 ---> ee0e15f2fccb
Step 5/7 : COPY app.py .
 ---> 20ae11a19a07
Step 6/7 : EXPOSE 5000
 ---> Running in 782a25daf844
 ---> Removed intermediate container 782a25daf844
 ---> 175a4021e54a
Step 7/7 : CMD ["python", "app.py"]
 ---> Running in 04ed612ab45f
 ---> Removed intermediate container 04ed612ab45f
 ---> 5557786ca37d
Successfully built 5557786ca37d
Successfully tagged simple-web-app:latest
cse@cse-B760M-H:~/docker-simple-web$
```

```
docker images
```

```
cse@cse-B760M-H:~/docker-simple-web$ docker images
REPOSITORY          TAG         IMAGE ID       CREATED              SIZE
simple-web-app      latest      5557786ca37d   About a minute ago   134MB
hello-world-image   latest      b7e187b4b9ab   51 minutes ago       77.9MB
python              3.10-slim   988b1bc80158   8 days ago           122MB
ubuntu              22.04       65c77cbc27c2   5 weeks ago          77.9MB
cse@cse-B760M-H:~/docker-simple-web$
```

Step 6: Run the Docker Container

```
docker run -d -p 5000:5000 simple-web-app
```

```
cse@cse-B760M-H:~/docker-simple-web$ docker run -d -p 5000:5000 simple-web-app
b116a6238394798da0a121540a2d02113926ee65b46816d1dcb97881cac61e0c
cse@cse-B760M-H:~/docker-simple-web$
```

Note:

- *`-d` → detached mode*
- *`-p 5000:5000` maps container port to host port*

Step 7: Verify Container is Running

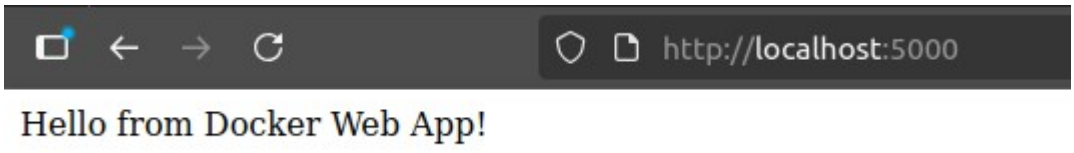
```
docker ps
```

```
cse@cse-B760M-H:~/docker-simple-web$ docker ps
CONTAINER ID   IMAGE            COMMAND            CREATED          STATUS          PORTS                                          NAMES
b116a6238394   simple-web-app   "python app.py"    37 seconds ago   Up 36 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp    competent_cray
```

Step 8: Access the Web Application

Open browser and visit:

http://localhost:5000



Hello from Docker Web App!

Step 9: Stop the Container (Cleanup)

```
docker stop <container_id>
```

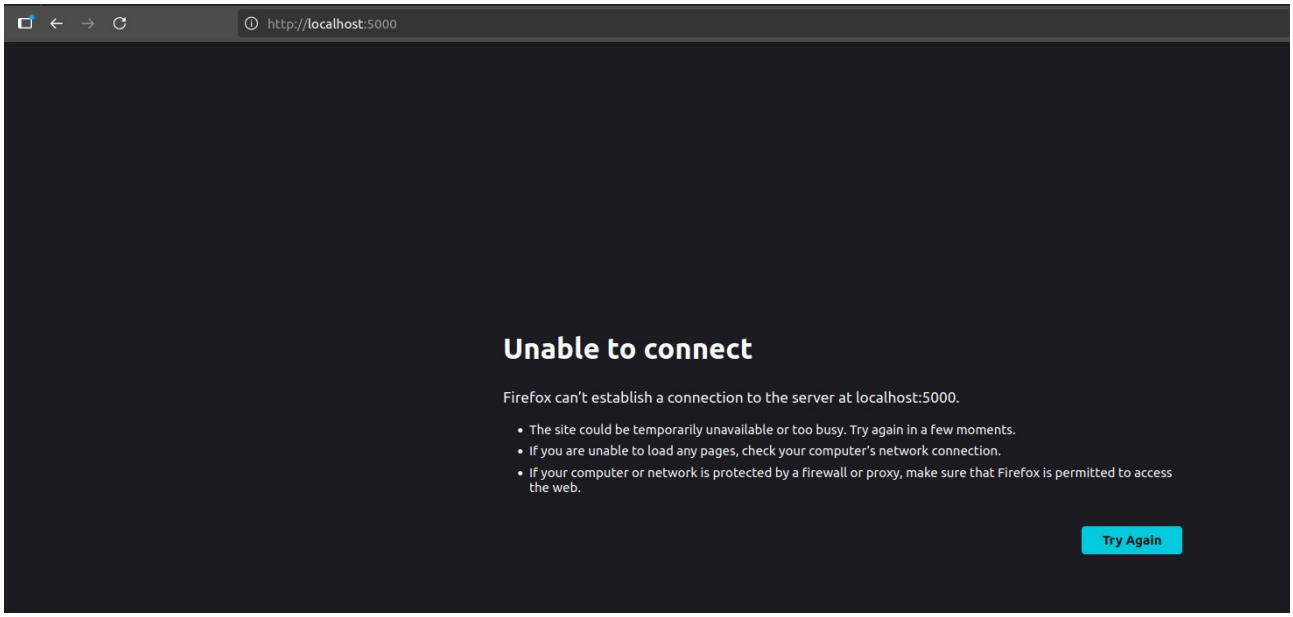
```
cse@cse-B760M-H:~/docker-simple-web$ docker stop b116a6
b116a6
cse@cse-B760M-H:~/docker-simple-web$
```

```
docker ps
```

```
cse@cse-B760M-H:~/docker-simple-web$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS   PORTS   NAMES
cse@cse-B760M-H:~/docker-simple-web$
```

```
docker ps -a
```

```
cse@cse-B760M-H:~/docker-simple-web$ docker ps -a
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS                      PORTS     NAMES
b116a6238394   simple-web-app     "python app.py"          5 minutes ago    Exited (137) 2 minutes ago            competent_cray
72326bcd11bc   hello-world-image  "echo 'Hello World'"     57 minutes ago   Exited (0) 57 minutes ago             eloquent_dirac
cse@cse-B760M-H:~/docker-simple-web$
```

Step 10: Ensure that the Web Application is Not Running

Go back to the browser and refresh the page.